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ROBUST CONTROL OF LINEAR SYSTEMS WITH SWITCHED ACTUATORS SUBJECTED TO DWELL-TIME CONSTRAINTS

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*"I am and always will be the optimist.
The hoper of far-flung hopes, and the dreamer of improbable dreams."*
— The Eleventh Doctor

Resumo

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o systema à trajetória cíclica alvo.

Abstract

Switched actuators are difficult to handle because of some characteristic constraints of their nature. but they are efficient in numerous engineering applications, being applied, for example, in attitude control systems for space vehicles. Thus, knowing how to use this type of actuator is important and necessary. The main problem to be solved in this thesis is the design of a controller that guarantees the robustness of a system with switched actuators subjected to time constraints that limit the response window of their commands. In this approach, a limit cycle target trajectory search is performed and then the error of the states compared to the target trajectory is calculated. Then a predictive control technique is applied to correct the error and get the system to the cyclic target trajectory

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List of Abbreviations and Acronyms

CTq	computed torque
DC	direct current
EAR	Equação Algébrica de Riccati
GDL	graus de liberdade
ISR	interrupção de serviço e rotina
LMI	linear matrices inequalities
MIMO	multiple input multiple output
PD	proporcional derivativo
PID	proporcional integrativo derivativo
PTP	point to point
UARMII	Underactuated Robot Manipulator II
VSC	variable structure control

List of Symbols

a	Distância
$\mathbf{a}$	Vetor de distâncias
$\mathbf{e}_j$	Vetor unitário de dimensão n e com o j-ésimo componente igual a 1
$\mathbf{K}$	Matriz de rigidez
m_1	Massa do cumpim
δ_{k-k_f}	Delta de Kronecker no instante k_f

Contents

1 Introduction

1.1 Motivation

The dynamics of switched affine systems can be described by a state equation of the form $\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, t \in \mathbb{R}_+$. Where $x(t) \in \mathbb{R}^n$ is the state vector and $\sigma(t) \in \{1, 2, ..., N\}$ is the switching state, which is associated to the matrix $A_{\sigma(t)} \in \mathbb{R}^{n\times n}$ and the vector $b_{\sigma(t)} \in \mathbb{R}^n$ at each time t.

The use of switched affine systems to solve problems in engineering can be seen in fields like biochemical reactions, where Parise *et al.* (2018) uses switched affine systems to represent the moments of evolution in a generic biochemical reaction network; urban traffic, where Hajiahmadi *et al.* (2016) approaches the urban traffic control; power converters, where Benmiloud *et al.* (2019) also uses this approach and other areas where the behavior of the system can be described as a switched affine system.

There is extensive research in this class of controllers, and most of them solve the problem by driving the system states to the neighborhood of the target states, not considering the trajectory behavior of the system. This can reduce the performance of the system when the system is affected by the trajectory of the limit cycle in steady operation.

This problem has been the target of many studies, Benmiloud *et al.* (2019) uses a hybrid Poincaré map approach to stabilize a switched affine system into the desired limit cycle and uses as an example a DC-DC converter. by Egidio *et al.* (2020b), which proposes a control design of a state-dependent switching function for discrete-time switched affine systems to drive the system to the desired limit cycle that is defined by design associated with the expected performance of the system. The solution was applied in an academic numerical example and a theoretical DC-DC converter. In Egidio *et al.* (2020a) the author expanded the work to continuous-time switched affine systems.

Some studies form the basis for the present work:

Vieira (2013) solved the problem for a particular class of affine systems. Here the controller drove the system to a target state using MILP (Mixed-Integer Linear Programming) and the cyclic trajectory was a consequence of the states as the system approached the target.

Patino *et al.* (2010) defined the target trajectory using nonlinear programming and used this target trajectory in the predictive control in terms of errors.

Marcolino (2018) proposed a solution with a controller law that drives the switched affine system with dwell-time constraints to a target cyclic trajectory. In this work, the problem is addressed in two stages: first, the problem is interpreted as a convex optimization where the author can find a targeted cyclic trajectory that minimizes a cost function associated with the system performance; then a second part that uses the target trajectory as input and computes perturbation at the switching instants of the reference control signal using predictive control.

1.2 Contributions of the present work

This present work proposes to expand the solution from)matheus2018 and provide a broader solution that can be used for systems that have their dynamics changing over time.

In a system that can be represented by a switched affine system with the form

$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}$

there are cases in which the dynamic response of the system can change depending on the operation mode. This can be represented in switched affine systems with proper matrices $A_{\sigma(t)}$ and $b_{\sigma(t)}$ for each operation mode. With the current implementation of Marcolino (2018) considering an invariant system during its operation, there is a need for expanding the solution so it can represent more systems and solve more engineering problems that can be expressed as switched affine systems.

1.3 Outline of the remaining chapters

The remaining of this text is organized as follows. Chapter 2 describes the problem of finding a target periodic trajectory for the given switched affine system. Chapter 3 proposes then a predictive control law that drives the system to the target cyclic trajectory. Finally, chapter 4 brings some remarks and considerations of the work.

2 Determination of Target Periodic Trajectories

Devemos encontrar uma trajetoria ciclica que respeite todos as restricoes do sistema e minimize a distancia em torno do valor de referencia?

Funcao custo:

$$J(\tau^{\infty}, \Omega^{\infty}, s^{\infty}) = \min_{\tau,\Omega,s} \int_0^{T_p} [x(t) - \bar{x}^{\infty}]^T Q[x(t) - \bar{x}^{\infty}]dt \qquad (2.1)$$

As entradas para a otimizacao são:

- Lista de modos de operacao
- Lista de matriz A associada ao modo de operacao
- Lista de matriz B associada ao modo de operacao
- Matriz C
- Matriz D
- Lista de valores de controle em cada modo
- Valor de referencia dos estados (xref)
- Tempo maximo de trajetoria ciclica (Tpmax)
- Numero maximo de estados
- Peso dos estados na funcao custo (Q)

Em que $Q = Q^T > 0$ e T_p é o período do ciclo.

A condição inicial é determinada de forma que o ultimo ponto é coincidente com o primeiro ponto (dada a sequencia de modos e instantes de chaveamento)

$$x(0) = (I - F_N F_{N-1} \dots F_1)^{-1} c \tag{2.2}$$

$$c = F_N F_{N-1} \dots F_2 G_1 u_1 + F_N F_{N-1} \dots F_3 G_2 u_2 + \dots + F_N G_{N-1} + G_N u_N \tag{2.3}$$

O resultado da otimizacao é τ^∞ e x_0

Exemplos:

Integrador (Matheus): Nao há otimizacao aqui, usamos a mesma trajetoria do Matheus para que tivessemos uma comparacao entre os resultados das simulacoes.

Patino, aqui estou usando um otimizador para encontrar a trajetoria de referencia.

Resultados:

Descrever resultados trajetorias Patino 1 e 2

TODO: FAZER SIMULACAO COMPLETA DE CADA CASO

2.1 Statement of the problem

2.2 Optimization

description of the optimization for finding the target trajectory

2.3 Calculating x_r

$x_0 \in \Re$

$$x(t) = e^{At} x(0) + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau \tag{2.4}$$

Modos:

$$\begin{aligned}
[0 - t_1] &: \dot{x} = A_1 x + B_1 u_1 \\
[t_1 - t_2] &: \dot{x} = A_2 x + B_2 u_2 \\
&\vdots \\
[t_{N-1} - t_N] &: \dot{x} = A_N x + B_N u_N
\end{aligned} \tag{2.5}$$

$$
\begin{aligned}
x(t_1) &= e^{A_1 t_1} x(0) + \int_0^{t_1} e^{A_1(t_1-\tau)} B_1 u_1 d\tau \\
x(t_1) &= e^{A_1 t_1} x(0) + \left[\int_0^{t_1} e^{A_1(t_1-\tau)} B_1 d\tau \right] u_1 \\
x(t_1) &= F_1 x(0) + G_1 u_1 \qquad (2.6)
\end{aligned}
$$

$$
\begin{aligned}
x(t_1) &= F_1 x(0) + G_1 u_1 \qquad (2.7) \\
x(t_2) &= F_2 x(1) + G_2 u_2 \qquad (2.8) \\
&\vdots \\
x(t_N) &= F_N x(N-1) + G_N u_N \qquad (2.9)
\end{aligned}
$$

$$
\begin{aligned}
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&\quad F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N) \\
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + c \qquad (2.10)
\end{aligned}
$$

Queremos $x(t_N) = x(0) = ?$

$$
\begin{aligned}
x(0) &= F_N F_{N-1} \dots F_1 x(0) + c \\
\underbrace{(I - F_N F_{N-1} \dots F_1)}_{\text{suposta invertível}} x(0) &= c \\
x(0) &= (I - F_N F_{N-1} \dots F_1)^{-1} c \qquad (2.11)
\end{aligned}
$$

em que

$$
\begin{aligned}
c = &F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N)
\end{aligned} \qquad (2.12)
$$

2.4 Application Examples

2.4.1 Variant Dynamic Circuit 1

2.4.2 Variant Dynamic Circuit 2

3 Predictive control: Preliminaries and notation

Assumesse que o controle e estados têm que pertencer ao seguinte conjuntos

$$\mathcal{U} = \{\mathbf{u} \in \mathbb{R}^p : S_u u \leq b_u\} \tag{3.1}$$

and

$$\mathcal{X} = \{x \in \mathbb{R}^n : S_x x \leq b_x\}, \mathcal{X}_f = \{x \in \mathbb{R}^n : S_f x \leq b_f\} \tag{3.2}$$

X_f é o máximo conjunto invariante e admissível (MAS) com respeito às restrições de estado e controle terminal sobre acao de uma forca de controle terminal

3.1 Cost Function

The solution, if it exists, will not be unique. Therefore, it is necessary to adopt a criterion that allows for choosing the best sequence of control actions to perform the task.

The control problem consists of bringing the states to the origin, so we will employ a cost in the form

$$J = \|x(N)\|^2_{P_f} + \sum_{k=0}^{N-1} \left[\|x(k)\|^2_Q + \|u(k)\|^2_R\right] \tag{3.3}$$

The expression for J consists of a terminal cost term and a trajectory cost term.

With P_f, Q, R being symmetric and positive-definite, with P_f chosen so that the terminal cost term $\|\hat{x}(k+N|k) - \bar{x}\|^2_{P_f}$ corresponds to the value function of the infinite horizon Discrete-Time Linear Quadratic Regulator (DLQR).

The cost function can be rewritten using the notation:

$$\mathbf{u} = \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \in \mathbb{R}^{pN}, \mathbf{x} = \begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix} \in \mathbb{R}^{nN} \tag{3.4}$$

and then the sum equation

$$\sum_{k=0}^{N-1} \|u(k)\|_R^2 = u^T(0)Ru(0) + u^t(1)Ru(1) + \cdots + u^T(N-1)Ru(N-1) \tag{3.5}$$

becomes

$$\underbrace{\begin{bmatrix} u^T(0) & u^T(1) & \cdots & u^T(N-1) \end{bmatrix}}_{\mathbf{u}^T} \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} \underbrace{\begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix}}_{\mathbf{u}} \tag{3.6}$$

Using the notation

$$\mathbf{R} = \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} = diag_N(R) \tag{3.7}$$

we get to

$$\sum_{k=0}^{N-1} \|u(k)\|_R^2 = \|\mathbf{u}\|_{\mathbf{R}}^2 \tag{3.8}$$

In a similar way, we can write

$$
\begin{aligned}
\|x(N)\|_{P_f}^2 + \sum_{k=0}^{N-1} \|x(k)\|_Q^2 &= x^T(0)Qx(0) \\
+x^T(1)Qx(1) + \cdots + x^T(N-1)Qx(N-1) + x^T(N)P_f x(N) \\
&= x^T(0)Qx(0) + \\
\underbrace{\begin{bmatrix} x^T(1) & x^T(2) & \cdots & x^T(N) \end{bmatrix}}_{\mathbf{x}^T} &\begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} \underbrace{\begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix}}_{\mathbf{x}}
\end{aligned}
\tag{3.9}
$$

with the notation

$$
Q = \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} = \begin{bmatrix} diag_{N-1}(Q) & 0 \\ 0 & P_f \end{bmatrix} \in \mathbb{R}^{nN \times nN} \tag{3.10}
$$

as

$$
\|x(N)\|_f^2 + \sum_{k=0}^{N-1} \|x(k)\|_Q^2 = \|x(0)\|_Q^2 + \|\mathbf{x}\|_{\mathbf{Q}}^2 \tag{3.11}
$$

So, the cost function can be written in the form

$$
J(\mathbf{x}, \mathbf{u}) = \|x(0)\|_Q^2 + \|\mathbf{x}\|_{\mathbf{Q}}^2 + \|\mathbf{u}\|_{\mathbf{R}}^2 \tag{3.12}
$$

3.2 Restrictions

The restrictions are introduced aiming to drive the states of the system to the origin while solving

$$
\min_{\mathbf{x} \in \mathbb{R}^{nN}, \mathbf{u} \in \mathbb{R}^{pN}} \mathbf{J}(\mathbf{x}, \mathbf{u}) \tag{3.13}
$$

respecting constraints applied on

$$\mathbf{u} = \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \in \mathcal{U} \times \mathcal{U} \times \cdots \times \mathcal{U} = \mathcal{U}^N \tag{3.14}$$

$$\mathbf{x} = \begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix} \in \mathcal{X} \times \mathcal{X} \times \cdots \times \mathcal{X}_f = \mathcal{X}^{N-1} \times \mathcal{X}_f \tag{3.15}$$

The constraint on $\mathbf{u}$ can be written as

$$\begin{bmatrix} S_u & 0 & \cdots & 0 \\ 0 & S_u & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_u \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \leq \begin{bmatrix} b_u \\ b_u \\ \vdots \\ b_u \end{bmatrix} \tag{3.16}$$

or, in a shorter form

$$\mathbf{S_u u} \leq \mathbf{b_u} \tag{3.17}$$

with

$$\mathbf{S_u} = diag_N(S_u), \mathbf{b_u} = col_N(b_u) \tag{3.18}$$

Similarly, the constraints on $\mathbf{x}$ can be written as

$$\begin{bmatrix} S_x & 0 & \cdots & 0 \\ 0 & S_x & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_x \end{bmatrix} \begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix} \leq \begin{bmatrix} b_x \\ b_x \\ \vdots \\ b_x \end{bmatrix} \tag{3.19}$$

or, in short

$$\mathbf{S_x x} \leq \mathbf{b_x} \tag{3.20}$$

with

$$\mathbf{S_x} = \begin{bmatrix} diag_{N-1}(S_x) & 0 \\ 0 & S_f \end{bmatrix}, \mathbf{b_x} = \begin{bmatrix} col_{N-1}(b_x) \\ b_f \end{bmatrix} \tag{3.21}$$

3.3 Prediction Equation

Let's express the relationship between the vectors

$$\mathbf{u} = \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix} \tag{3.22}$$

based on the state equation

$$x(k+1) = Ax(k) + Bu(k) \tag{3.23}$$

and initial condition $x(0)$.

The solution for the state equation is:

$$x(k) = A^k x(0) + \sum_{i=0}^{k-1} A^{k-1-i} Bu(i) \tag{3.24}$$

and it can be written for $k = 1, 2, ..., N$ as

$$x(1) = Ax(0) + Bu(0) \tag{3.25}$$
$$x(2) = A^2 x(0) + ABu(0) + Bu(1) \tag{3.26}$$
$$\vdots \tag{3.27}$$
$$x(N) = A^N x(0) A^{N-1} Bu(0) + \cdots + ABu(N-2) + Bu(N-1) \tag{3.28}$$

or

$$\begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} x(0) + \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \tag{3.29}$$

This can be written in a shorter form as

$$\mathbf{x} = \mathbf{A}x(0) + \mathbf{Bu} \tag{3.30}$$

with

$$\mathbf{A} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}, \mathbf{B} = \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \tag{3.31}$$

==================================

// NOTE: mudar funcao custo para '$\bar{x} = 0$'

==================================

Tendo em vista o vínculo

$$\mathbf{x} = \mathbf{A}x(0) + \mathbf{Bu} \tag{3.32}$$

pode-se reescrever a função de custo como

$$J(\mathbf{x}, \mathbf{u}) = \|x(0) - \bar{x}\|_Q^2 + \overbrace{\|\mathbf{x} - col_N(\bar{x})\|_{\mathbf{Q}}^2}^{\square} + \overbrace{\|\mathbf{u} - col_N(\bar{u})\|_{\mathbf{R}}^2}^{\bigcirc} \tag{3.33}$$

$$\begin{aligned} \square &= \|\mathbf{x} - col_N(\bar{x})\|_{\mathbf{Q}}^2 && (3.34) \\ &= \left[\mathbf{A}x(0) + \mathbf{Bu} - col_N(\bar{x})\right]^T \mathbf{Q} \left[\mathbf{A}x(0) + \mathbf{Bu} - col_N(\bar{x})\right] && (3.35) \\ &= \left[\mathbf{A}x(0) - col_N(\bar{x}) + \mathbf{Bu}\right]^T \mathbf{Q} \left[\mathbf{A}x(0) - col_N(\bar{x}) + \mathbf{Bu}\right] && (3.36) \\ &= \|\mathbf{A}x(0) - col_N(\bar{x})\|_{\mathbf{Q}}^2 + \left[\mathbf{A}x(0) - col_N(\bar{x})\right]^T \mathbf{QBu} \\ &\quad + \mathbf{u}^T\mathbf{B}^T\mathbf{Q}\left[\mathbf{A}x(0) - col_N(\bar{x})\right] + \mathbf{u}^T\mathbf{B}^T\mathbf{QBu} && (3.37) \end{aligned}$$

$$\begin{aligned} \bigcirc &= \|\mathbf{u} - col_N(\bar{u})\|_{\mathbf{R}}^2 && (3.38) \\ &= \left[\mathbf{u} - col_N(\bar{u})\right]^T \mathbf{R} \left[\mathbf{u} - col_N(\bar{u})\right] && (3.39) \\ &= \mathbf{u}^T\mathbf{Ru} - \mathbf{u}^T\mathbf{R}\left[col_N(\bar{u})\right] - \left[col_N(\bar{u})\right]^T \mathbf{Ru} + \|col_N(\bar{u})\|_{\mathbf{R}}^2 && (3.40) \end{aligned}$$

substituindo $\square$ and $\bigcirc$ de volta na equacao de custo e reorganizando os termos

$$J(\mathbf{x},\mathbf{u}) = \|x(0)-\bar{x}\|_Q^2 + \|\mathbf{A}x(0) - col_N(\bar{x})\|_{\mathbf{Q}}^2 + \|col_N(\bar{u})\|_{\mathbf{R}}^2 \tag{3.41}$$

$$+\left[\mathbf{A}x(0) - col_N(\bar{x})\right]^T \mathbf{QBu} + \mathbf{u}^T\mathbf{B}^T\mathbf{Q}\left[\mathbf{A}x(0) - col_N(\bar{x})\right] \tag{3.42}$$

$$-\mathbf{u}^T\mathbf{R}\left[col_N(\bar{u})\right] - \left[col_N(\bar{u})\right]^T\mathbf{Ru} + \mathbf{u}^T(\mathbf{R}+\mathbf{B}^T\mathbf{QB})\mathbf{u} \tag{3.43}$$

O termo constante será denotado por c_0

Lembrando que a matriz $\mathbf{Q}$ é simétrica, tem-se

$$[\mathbf{A}x(0) - col_N(\bar{x})]^T\,\mathbf{QBu} = \left\{\mathbf{u}^T\mathbf{B}^T\mathbf{Q}\,[\mathbf{A}x(0) - col_N(\bar{x})]\right\}^T \tag{3.44}$$

Como esses termos são escalares, conclui-se que são iguais entre si. Logo:

$$\begin{gathered}[\mathbf{A}x(0) - col_N(\bar{x})]^T\,\mathbf{QBu} + \mathbf{u}^T\mathbf{B}^T\mathbf{Q}\,[\mathbf{A}x(0) - col_N(\bar{x})] \\ = 2\,[\mathbf{A}x(0) - col_N(\bar{x})]^T\,\mathbf{QBu}\end{gathered} \tag{3.45}$$

De forma similar:

$$-\mathbf{u}^T\mathbf{R}\,[col_N(\bar{u})] - [col_N(\bar{u})]^T\,\mathbf{Ru} = -2\,[col_N(\bar{u})]^T\,\mathbf{Ru} \tag{3.46}$$

Finalmente, pode-se escrever

$$\begin{gathered}J(\mathbf{u}) = \mathbf{u}^T(\mathbf{B}^T\mathbf{QB}+\mathbf{R})\mathbf{u} \\ +2\left\{[\mathbf{A}x(0) - col_N(\bar{x})]^T\,\mathbf{QB} - [col_N(\bar{u})]^T\,\mathbf{R}\right\}\mathbf{u} = c_0\end{gathered} \tag{3.47}$$

com

$$c_0 = \|x(0)-\bar{x}\|_Q^2 + \|\mathbf{A}x(0) - col_N(\bar{x})\|_{\mathbf{R}}^2 + \|col_N(\bar{u})\|_{\mathbf{R}}^2 \tag{3.48}$$

Note-se que passaremos a usar a notação $J(\mathbf{u})$ em lugar de $J(\mathbf{x},\mathbf{u})$, uma vez que $\mathbf{x}$ foi eliminada da expressão do custo.

Para terminar a reformulação do problema de otimização em termos de $\mathbf{u}$, deve-se considerar o vínculo entre $\mathbf{x}$ e $\mathbf{u}$ nas **restrições**

3.4 Problema de Otimização

O problema de otimização a ser resolvido pode ser expresso como

$$\min_{\mathbf{x}\in\mathbb{R}^{nN},\mathbf{u}\in\mathbb{R}^{pN}} J(x,u) \tag{3.49}$$

sendo

$$J(\mathbf{x},\mathbf{u}) = \|x(0)-\bar{x}\|_Q^2 + \|\mathbf{x} - col_N(\bar{x})\|_{\mathbf{Q}}^2 + \|\mathbf{u} - col_N(\bar{u})\|_{\mathbf{R}}^2 \tag{3.50}$$

com

$$\mathbf{u}(k) = \begin{bmatrix} u(k|k) \\ u(k+1|k) \\ \vdots \\ u(k+N-1|k) \end{bmatrix} \in \mathbb{R}^{pN}, \mathbf{x}(k) = \begin{bmatrix} x(k+1|k) \\ x(k+2|k) \\ \vdots \\ x(k+N|k) \end{bmatrix} \in \mathbb{R}^{nN}, \tag{3.51}$$

sujeitas a

$$\mathbf{S_x x}(k) \leq \mathbf{b_x} \tag{3.52}$$

$$\mathbf{S_u u}(k) \leq \mathbf{b_u} \tag{3.53}$$

Os vetores $\mathbf{x}(k)$ e $\mathbf{u}(k)$ estão vinculados pela seguinte **equação de precição**:

$$\mathbf{x}(k) = \mathbf{A}x(k) + \mathbf{B}\mathbf{u}(k) \tag{3.54}$$

Com isso, o custo pode ser reescrito como

$$J(\mathbf{u}(k)) = \mathbf{u}(k)^T(\mathbf{B}^T\mathbf{QB} + \mathbf{R})\mathbf{u}(k)$$
$$2\left\{[\mathbf{A}x(k) - col_N(\bar{x})]^T\mathbf{QB} - [col_N(\bar{u})]^T\mathbf{R}\right\}\mathbf{u}(k) + c_0$$

com

$$c_0 = \|x(k)-\bar{x}\|_Q^2 + \|\mathbf{A}x(k) - col_N(\bar{x})\|_{\mathbf{Q}}^2 + \|col_N(\bar{u})\|_{\mathbf{R}}^2$$

e as restrições passam a ser expressas na forma

$$\mathbf{Su}(k) \leq \mathbf{b}$$

com

$$\mathbf{S} = \begin{bmatrix} \mathbf{S_x B} \\ \mathbf{S_u} \end{bmatrix}, \mathbf{b} = \begin{bmatrix} \mathbf{b_x} - \mathbf{S_x A} x(k) \\ \mathbf{b_u} \end{bmatrix}$$

Tendo-se obtido a soluçao $\mathbf{u}(k)$ para o problema $\mathbb{P}\Big(x(k)\Big)$, o controle a ser aplicado à planta é dado por

$$u(k) = \hat{u}^*(k|k) \tag{3.55}$$

com $\hat{u}^*(k|k)$ correspondendo aos p primeiros elementos de $\hat{\mathbf{u}}^*(k)$

4 Predictive Control for Stabilization of Periodic Trajectories

4.1 Statement of the problem

Considering a plant of an affine switched system described by:

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)} \tag{4.1}$$

The control action applied in the system is in the form of perturbations in each operation mode switching time. By allowing the control the a system using predictive control, we need to find a linear equation that relates the switching times and the resulting system at the end of one cycle.

4.1.1 Cost Function

$$J = \|e[N_p]\|^2_{P_f} + \sum_{j=0}^{N_p-1} \|e[j]\|^2_Q + \|\delta t[j]\|^2_R \tag{4.2}$$

$$e(t_{j,N}) = \Phi e(t_{j,0}) + \Gamma \delta t[j] \tag{4.3}$$

em que $Q = Q^T > 0$, $\rho > 0$

$$\delta t = [t_{j,1} - \bar{t}_{j,1}, t_{j,2} - \bar{t}_{j,2}, ...]^T \tag{4.4}$$

4.1.2 Dwell-time Constraints

// TODO: colocar as Dwell-time Constraints (3.1.2), dwell time

4.2 Linearization procedure

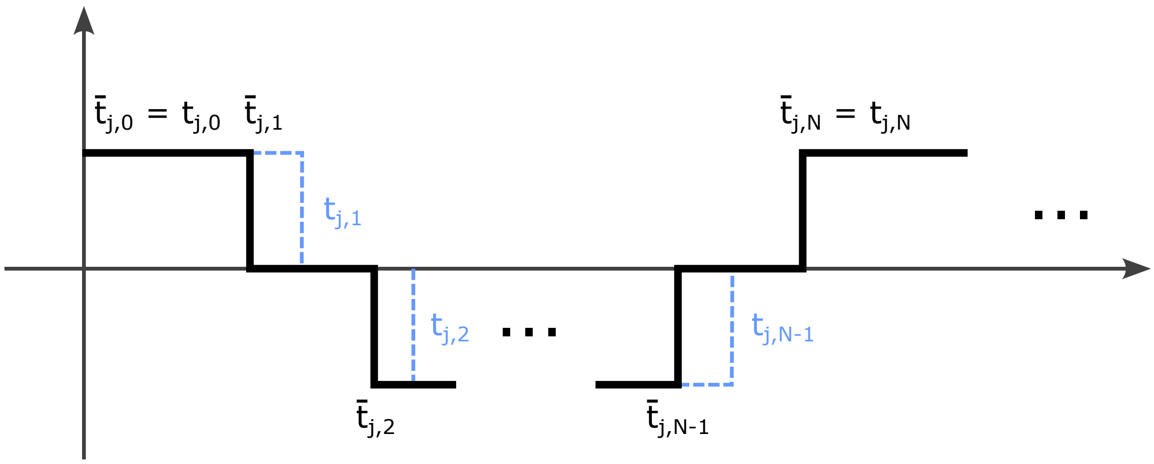

FIGURE 4.1 – Instantes de chaveamento

The solution for the state equation (4.1) at time t_1, starting from the initial condition $x(t_0)$, is of the form

$$x(t_1) = e^{A_0(t_1 - t_0)}x(t_0) + \int_{t_0}^{t_1} e^{A_0(t_1-\tau)}B_0 d\tau \tag{4.5}$$

$$\bar{x}(\bar{t}_1) = e^{A_0(\bar{t}_1 - \bar{t}_0)}\bar{x}(\bar{t}_0) + \int_{\bar{t}_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0 d\tau \tag{4.6}$$

subtracting (4.6) from (4.5) we get

$$\begin{aligned} x(t_1) - \bar{x}(\bar{t}_1) = & \overbrace{e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0)}^{\square} \\ & \underbrace{+ \int_{t_0}^{t_1} e^{A_0(t_1-\tau)}B_0 d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0 d\tau}_{\bigcirc} \end{aligned} \tag{4.7}$$

expanding (4.7) first in $\square$:

$$\begin{aligned} \square &= e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= e^{A_0(\bar{t}_1+\delta t_1-t_0)}x(t_0) - e^{A_0(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= e^{A_0(\bar{t}_1-t_0)}e^{\delta t_1}x(t_0) - e^{A_0(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= \bar{F}_0 e^{\delta t_1}x(t_0) - \bar{F}_0\bar{x}(\bar{t}_0) \\ &= \bar{F}_0(I + A_0\delta t_1)x(t_0) - \bar{F}_0\bar{x}(\bar{t}_0) \\ &= \bar{F}_0[x(t_0) - \bar{x}(t_0)] + \bar{F}_0 A_0 \delta t_1 \end{aligned} \tag{4.8}$$

and then in $\bigcirc$:

$$
\begin{aligned}
\bigcirc &= \int_{t_0}^{t_1} e^{A_0(t_1-\tau)}B_0 d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0 d\tau \\
&= \int_{t_0}^{\bar{t}_1+\delta t_1} e^{A_0(\bar{t}_1+\delta t_1-\tau)}B_0 d\tau - \int_{t_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0 d\tau \\
&= e^{A_0\delta t_1}\left[\int_{t_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0 d\tau + \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_0(t_{r_1}-\tau)}B_0 d\tau\right] - \int_{t_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0 d\tau \\
&= e^{A_0\delta t_1}[\bar{G}_0 + B_0\delta t_1] - \bar{G}_0 \\
&= (I + A_0\delta t_1)[\bar{G}_0 + B_0\delta t_1] - \bar{G}_0 \\
&= \cancel{\bar{G}_0} + B_0\delta t_1 + A_0\bar{G}_0\delta t_1 + \cancel{A_0B_0\delta t_1\delta t_1} - \cancel{\bar{G}_0} \\
&= A_0\bar{G}_0\delta t_1 + B_0\delta t_1
\end{aligned} \tag{4.9}
$$

the resulting equations (4.8) and (4.9) are replaced back in (4.7) to complete the difference equation

$$
\boxed{x(t_1) - \bar{x}(\bar{t}_1) = \bar{F}_0[x(t_0) - \bar{x}(t_0)] + [\bar{F}_0A_0x(t_0) + A_0\bar{G}_0 + B_0]\delta t_1} \tag{4.10}
$$

Repeating the steps for $x(t_2)$ we have

$$
x(t_2) = e^{A_1(t_2-t_1)}x(t_1) + \int_{t_1}^{t_2} e^{A_1(t_2-\tau)}B_1 d\tau \tag{4.11}
$$

$$
\bar{x}(\bar{t}_2) = e^{A_1(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) + \int_{\bar{t}_1}^{\bar{t}_2} e^{A_1(\bar{t}_2-\tau)}B_1 d\tau \tag{4.12}
$$

subtracting (4.12) from (4.11)

$$
\begin{aligned}
x(t_2) - \bar{x}(\bar{t}_2) = &\overbrace{e^{A_1(t_2-t_1)}x(t_1) - e^{A_1(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1)}^{\square} \\
&\underbrace{+\int_{t_1}^{t_2} e^{A_1(t_2-\tau)}B_1 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_1(\bar{t}_2-\tau)}B_1 d\tau}_{\bigcirc}
\end{aligned} \tag{4.13}
$$

expanding $\square$ in (4.13):

$$
\begin{aligned}
\square &= e^{A_1(\bar{t}_2+\delta t_2-\bar{t}_1-\delta t_1)}x(t_1) - e^{A_1(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\
&= e^{A_1(\bar{t}_2-\bar{t}_1)}e^{A_1(\delta t_2-\delta t_1)}x(t_1) - e^{A_1(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\
&= \bar{F}_1[I + A_1(\delta t_2 - \delta t_1)]x(t_1) - \bar{F}_1\bar{x}(\bar{t}_1) \\
&= \bar{F}_1[x(t_1) - \bar{x}(\bar{t}_1)] + \bar{F}_1A_1x(t_1)\delta t_2 - \bar{F}_1A_1x(t_1)\delta t_1
\end{aligned} \tag{4.14}
$$

and expanding ◯:

$$
\begin{aligned}
\bigcirc &= \int_{t_1}^{t_2} e^{A_1(t_2-\tau)}B_1 d\tau - \int_{\bar t_1}^{\bar t_2} e^{A_1(\bar t_2-\tau)}B_1 d\tau \\
&= \int_{\bar t_1+\delta t_1}^{\bar t_2+\delta t_2} e^{A_1(\bar t_2+\delta t_2-\tau)}B_1 d\tau - \int_{\bar t_1}^{\bar t_2} e^{A_1(\bar t_2-\tau)}B_1 d\tau \\
&= e^{A_1\delta t_2}\left[\int_{\bar t_1}^{\bar t_2} e^{A_1(\bar t_2-\tau)}B_1 d\tau - \int_{\bar t_1}^{\bar t_1+\delta t_1} e^{A_1(\bar t_2-\tau)}B_1 d\tau + \int_{\bar t_2}^{\bar t_2+\delta t_2} e^{A_1(\bar t_2-\tau)}B_1 d\tau\right] \\
&\quad - \int_{\bar t_1}^{\bar t_2} e^{A_1(\bar t_2-\tau)}B_1 d\tau \\
&= (I + A_1\delta t_2)\left[\bar G_1 - \int_{\bar t_1}^{\bar t_1+\delta t_1} e^{A_1(\bar t_2-\tau)}B_1 d\tau + \int_{\bar t_2}^{\bar t_2+\delta t_2} e^{A_1(\bar t_2-\tau)}B_1 d\tau\right] - \bar G_1 \\
&= (I + A_1\delta t_2)\left[\bar G_1 - e^{A_1(\bar t_2-\bar t_1)}B_1\delta t_1 + B_1\delta t_2\right] - \bar G_1 \\
&= (I + A_1\delta t_2)\left[\bar G_1 - \bar F_1 B_1\delta t_1 + B_1\delta t_2\right] - \bar G_1 \\
&= \cancel{\bar G_1} - \bar F_1 B_1\delta t_1 + B_1\delta t_2 + A_1\bar G_1\delta t_2 - \cancel{\bar G_1} \\
&= A_1\bar G_1\delta t_2 + B_1\delta t_2 - \bar F_1 B_1\delta t_1
\end{aligned} \tag{4.15}
$$

and replacing (4.14) and (4.15) back into equation (4.13) we get to the difference equation for $x(t_2) - \bar x(\bar t_2)$

$$
\begin{aligned}
x(t_2) - \bar x(\bar t_2) = &\bar F_1[x(t_1) - \bar x_1(\bar t_1)] \\
&- \bar F_1[A_1 x(t_1) + B_1]\delta t_1 \\
&+ [\bar F_1 A_1 \underbrace{x(t_1)}_{\triangle} + A_1\bar G_1 + B_1]\delta t_2
\end{aligned} \tag{4.16}
$$

from (4.16), △ can be expanded by replacing the resulting equation of (4.10)

$$\triangle = x(t_1) = \bar x(\bar t_1) + \bar F_0[x(t_0) - \bar x(t_0)] + [\bar F_0 A_0 x(t_0) + A_0\bar G_0 + B_0]\delta t_1 \tag{4.17}$$

note that when △ is replaced in eq. (4.16) it will produce a product with δt_1 and δt_2 like

$$x(t_1)\delta t_1 = \bar x(\bar t_1)\delta t_1 + \bar F_0\overbrace{[x(t_0) - \bar x(t_0)]\delta t_1}^{\square} + [\bar F_0 A_0 x(t_0) + A_0\bar G_0 + B_0]\overbrace{\delta t_1\delta t_1}^{\bigcirc}$$

both □ and ◯ are products of differences that produces a smaller number and they will be neglected in the first order aproximations

$$x(t_1)\delta t_1 \doteq \bar x(\bar t_1)\delta t_1 \tag{4.18}$$

replacing (4.10) and (4.17) in (4.16) it follows that

$$
\begin{aligned}
x(t_2) - \bar{x}(\bar{t}_2) =& \bar{F}_1\{\bar{F}_0[x(t_0) - \bar{x}(t_0) + [\bar{F}_0 A_0 x(t_0) + A_0\bar{G}_0 + B_0]\delta t_1]\} \\
& - \bar{F}_1[A_1\bar{x}(t_1) + B_1]\delta t_1 \\
& + [\bar{F}_1 A_1\bar{x}(t_1) + A_1\bar{G}_1 + B_1]\delta t_2
\end{aligned}
$$

$$
\boxed{
\begin{aligned}
x(t_2) - \bar{x}(\bar{t}_2) =& \bar{F}_1\bar{F}_0[x(t_0) - \bar{x}(t_0)] \\
& + \bar{F}_1[\bar{F}_0 A_0 x(t_0) + A_0\bar{G}_0 - A_1\bar{x}(\bar{t}_1) + B_0 - B_1]\delta t_1 \\
& + [\bar{F}_1 A_1\bar{x}(\bar{t}_1) + A_1\bar{G}_1 + B_1]\delta t_2
\end{aligned}
}
\tag{4.19}
$$

Repeating the procedure for $x(t_3)$ we have

$$
x(t_3) = e^{A_2(t_3-t_2)}x(t_2) + \int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2 d\tau \tag{4.20}
$$

$$
\bar{x}(\bar{t}_3) = e^{A_2(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2) + \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)}B_2 d\tau \tag{4.21}
$$

subtracting (4.20) from (4.21)

$$
\begin{aligned}
x(t_3) - \bar{x}(\bar{t}_3) = & \overbrace{e^{A_2(t_3-t_2)}x(t_2) - e^{A_2(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2)}^{\square} \\
& + \underbrace{\int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)}B_2 d\tau}_{\bigcirc}
\end{aligned}
\tag{4.22}
$$

expanding $\square$ in (4.22):

$$
\begin{aligned}
\square &= e^{A_2(\bar{t}_3+\delta t_3-\bar{t}_2-\delta t_2)}x(t_2) - e^{A_2(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2) \\
&= e^{A_2(\bar{t}_3-\bar{t}_2)}e^{A_2(\delta t_3-\delta t_2)}x(t_2) - e^{A_2(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2) \\
&= \bar{F}_2[I + A_2(\delta t_3 - \delta t_2)]x(t_2) - \bar{F}_2\bar{x}(\bar{t}_2) \\
&= \bar{F}_2[x(t_2) - \bar{x}(\bar{t}_2)] + \bar{F}_2 A_2 x(t_2)\delta t_3 - \bar{F}_2 A_2 x(t_2)\delta t_2
\end{aligned}
\tag{4.23}
$$

and expanding $\bigcirc$:

$$
\begin{aligned}
\bigcirc &= \int_{t_2}^{t_3} e^{A_2(t_3-\tau)} B_2 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau \\
&= \int_{\bar{t}_2+\delta t_2}^{\bar{t}_3+\delta t_3} e^{A_2(\bar{t}_3+\delta t_3-\tau)} B_2 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau \\
&= e^{A_2\delta t_3}\left[\int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau\right] \\
&\quad - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau \\
&= (I + A_2\delta t_3)\left[\bar{G}_2 - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau\right] - \bar{G}_2 \\
&= (I + A_2\delta t_3)\left[\bar{G}_2 - e^{A_2(\bar{t}_3-\bar{t}_2)} B_2\delta t_2 + B_2\delta t_3\right] - \bar{G}_2 \\
&= (I + A_2\delta t_3)\left[\bar{G}_2 - \bar{F}_2 B_2\delta t_2 + B_2\delta t_3\right] - \bar{G}_2 \\
&= \cancel{\bar{G}_2} - \bar{F}_2 B_2\delta t_2 + B_2\delta t_3 + A_2\bar{G}_2\delta t_3 - \cancel{\bar{G}_2} \\
&= A_2\bar{G}_2\delta t_3 + B_2\delta t_3 - \bar{F}_2 B_2\delta t_2
\end{aligned} \tag{4.24}
$$

and then replacing (4.23) in (4.24) back in eq. (4.22) we get to the difference equation for $x(t_3) - \bar{x}(\bar{t}_3)$

$$
\begin{aligned}
x(t_3) - \bar{x}(\bar{t}_3) =& \bar{F}_2[x(t_2) - \bar{x}_2(\bar{t}_2)] \\
&- \bar{F}_2[A_2 x(t_2) + B_2]\delta t_2 \\
&+ [\bar{F}_2 A_2 x(t_2) + A_2\bar{G}_2 + B_2]\delta t_3
\end{aligned} \tag{4.25}
$$

from (4.19), $\triangle$ can be expanded the same way as in the past iteration and $x(t_2) - \bar{x}(\bar{t}_2)$ can be replaced in (4.25)

$$
\begin{aligned}
x(t_3) - \bar{x}(\bar{t}_3) =& \bar{F}_2\{\bar{F}_1\bar{F}_0[x(t_0) - \bar{x}(t_0)] \\
&+ \bar{F}_1[\bar{F}_0 A_0 x(t_0) + A_0\bar{G}_0 - A_1\bar{x}(\bar{t}_1) + B_0 - B_1]\delta t_1 \\
&+ [\bar{F}_1 A_1\bar{x}(t_1) + A_1\bar{G}_1 + B_1]\delta t_2\} \\
&- \bar{F}_2[A_2\bar{x}(t_2) + B_2]\delta t_2 \\
&+ [\bar{F}_2 A_2\bar{x}(t_2) + A_2\bar{G}_2 + B_2]\delta t_3
\end{aligned}
$$

after rearranging the terms, one arrives at

$$\boxed{\begin{aligned} x(t_3)-\bar{x}(\bar{t}_3)=&\bar{F}_2\bar{F}_1\bar{F}_0[x(t_0)-\bar{x}(t_0)]\\ &+\bar{F}_2\bar{F}_1[\bar{F}_0A_0x(t_0)+A_0\bar{G}_0-A_1\bar{x}(\bar{t}_1)+B_0-B_1]\delta t_1\\ &+\bar{F}_2[\bar{F}_1A_1\bar{x}(t_1)+A_1\bar{G}_1-A_2\bar{x}(\bar{t}_2)+B_1-B_2]\delta t_2\\ &+[\bar{F}_2A_2\bar{x}(t_2)+A_2\bar{G}_2+B_2]\delta t_3 \end{aligned}} \tag{4.26}$$

Now, there are enough iterations to deduce a pattern on each increment of the equation, so the equation for $x(t_i)$ can be deduced as

$$\boxed{\begin{aligned} x(t_i)-\bar{x}(\bar{t}_i)=&\bar{F}_{i-1}\bar{F}_{i-2}\dots\bar{F}_0[x(t_0)-\bar{x}(t_0)]\\ &+\bar{F}_{i-1}\bar{F}_{i-2}\dots\bar{F}_1[\bar{F}_0A_0x(t_0)+A_0\bar{G}_0-A_1\bar{x}(\bar{t}_1)+B_0-B_1]\delta t_1\\ &+\bar{F}_{i-1}\bar{F}_{i-2}\dots\bar{F}_2[\bar{F}_1A_1x(t_1)+A_1\bar{G}_1-A_2\bar{x}(\bar{t}_2)+B_1-B_2]\delta t_2\\ &\quad\vdots\\ &+\bar{F}_{i-1}[\bar{F}_{i-2}A_{i-2}\bar{x}(t_{i-2})+A_{i-2}\bar{G}_{i-2}-A_{i-1}\bar{x}(\bar{t}_{i-1})+B_{i-2}-B_{i-1}]\delta t_{i-1}\\ &+[\bar{F}_{i-1}A_{i-1}\bar{x}(t_{i-1})+A_{i-1}\bar{G}_{i-1}+B_{i-1}]\delta t_i \end{aligned}} \tag{4.27}$$

and then, for $x(t_N)$ the equation is as follows

$$\begin{aligned} x(t_N)-\bar{x}(\bar{t}_N)=&\bar{F}_{N-1}\bar{F}_{N-2}\dots\bar{F}_0[x(t_0)-\bar{x}(t_0)]\\ &+\bar{F}_{N-1}\bar{F}_{N-2}\dots\bar{F}_1\overbrace{[\bar{F}_0A_0x(t_0)+A_0\bar{G}_0-A_1\bar{x}(\bar{t}_1)+B_0-B_1]}^{\square}\delta t_1\\ &+\bar{F}_{N-1}\bar{F}_{N-2}\dots\bar{F}_2[\bar{F}_1A_1x(t_1)+A_1\bar{G}_1-A_2\bar{x}(\bar{t}_2)+B_1-B_2]\delta t_2\\ &\quad\vdots\\ &+\bar{F}_{N-1}[\bar{F}_{N-2}A_{N-2}\bar{x}(t_{N-2})+A_{N-2}\bar{G}_{N-2}-A_{N-1}\bar{x}(\bar{t}_{N-1})\\ &\qquad\qquad+B_{N-2}-B_{N-1}]\delta t_{N-1}\\ &+\underbrace{[\bar{F}_{N-1}A_{N-1}\bar{x}(t_{N-1})+A_{N-1}\bar{G}_{N-1}+B_{N-1}]}_{\bigcirc}\delta t_N \end{aligned} \tag{4.28}$$

The equation (4.28) can be developed further to remove the dependencies on the intermediary states $x(t_0), x(t_1), ..., x(t_{N-2})$. By expanding $\square$ and $\bigcirc$, the following identity holds

$$
\begin{aligned}
\square &= \bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1 \\
&= A_0 \bar{F}_0 x(t_0) + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1 \\
&= A_0 [\bar{x}(\bar{t}_1) - \bar{G}_0] + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1 \\
&= A_0 \bar{x}(\bar{t}_1) - \cancel{A_0 \bar{G}_0} + \cancel{A_0 \bar{G}_0} - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1 \\
&= (A_0 - A_1) \bar{x}(\bar{t}_1) + B_0 - B_1
\end{aligned} \tag{4.29}
$$

$$
\begin{aligned}
\bigcirc &= \bar{F}_{N-1} A_{N-1} \bar{x}(t_{N-1}) + A_{N-1} \bar{G}_{N-1} + B_{N-1} \\
&= A_{N-1} \bar{F}_{N-1} \bar{x}(t_{N-1}) + A_{N-1} \bar{G}_{N-1} + B_{N-1} \\
&= A_{N-1} [\bar{x}(\bar{t}_N) - \bar{G}_{N-1}] + A_{N-1} \bar{G}_{N-1} + B_{N-1} \\
&= A_{N-1} \bar{x}(\bar{t}_N) - \cancel{A_{N-1} \bar{G}_{N-1}} + \cancel{A_{N-1} \bar{G}_{N-1}} + B_{N-1} \\
&= A_{N-1} \bar{x}(\bar{t}_N) + B_{N-1}
\end{aligned} \tag{4.30}
$$

replacing (4.29) and (4.30) in (4.28)

$$
\boxed{
\begin{aligned}
x(t_N) - \bar{x}(\bar{t}_N) = \quad & \bar{F}_{N-1} \bar{F}_{N-2} \dots \bar{F}_0 [x(t_0) - \bar{x}(t_0)] \\
& + \bar{F}_{N-1} \bar{F}_{N-2} \dots \bar{F}_1 [(A_0 - A_1) \bar{x}(\bar{t}_1) + B_0 - B_1] \delta t_1 \\
& + \bar{F}_{N-1} \bar{F}_{N-2} \dots \bar{F}_2 [(A_1 - A_2) \bar{x}(\bar{t}_2) + B_1 - B_2] \delta t_2 \\
& \quad \vdots \\
& + \bar{F}_{N-1} [(A_{N-2} - A_{N-1}) \bar{x}(\bar{t}_{N-1}) + B_{N-2} - B_{N-1}] \delta t_{N-1} \\
& + [A_{N-1} \bar{x}(\bar{t}_N) + B_{N-1}] \delta t_N
\end{aligned}
} \tag{4.31}
$$

and renaming the terms of (4.31) to

$$e(t) = x(t) - \bar{x}(\bar{t}) \tag{4.32}$$

$$\Phi = \bar{F}_{N-1}\bar{F}_{N-2}\dots\bar{F}_0 \tag{4.33}$$

$$\begin{aligned}\Gamma = \Big[&\bar{F}_{N-1}\bar{F}_{N-2}\dots\bar{F}_1[(A_0 - A_1)\bar{x}(\bar{t}_1) + B_0 - B_1],\\ &\bar{F}_{N-1}\bar{F}_{N-2}\dots\bar{F}_2[(A_1 - A_2)\bar{x}(\bar{t}_2) + B_1 - B_2],\\ &\quad\vdots\\ &\bar{F}_{N-1}[(A_{N-2} - A_{N-1})\bar{x}(\bar{t}_{N-1}) + B_{N-2} - B_{N-1}]\Big]\end{aligned} \tag{4.34}$$

$$\delta t = [\delta t_1, \delta t_2, \dots, \delta t_N]^T \tag{4.35}$$

we get to the affine equation

$$\boxed{e(t_N) = \Phi e(t_0) + \Gamma\delta t} \tag{4.36}$$

falar que chegamos no final

$$e(t_{j,N}) = \Phi e(t_{j,0}) + \Gamma\delta t[j] \tag{4.37}$$

TODO: voltar para o problema inicial em que queriamos encontrar os tj,n

where $\bar{x}$ is calculated as

$$\bar{x}(\bar{t}_1) = F_0\bar{x}(\bar{t}_0) + G_0 \tag{4.38}$$

$$\bar{x}(\bar{t}_2) = F_1\bar{x}(\bar{t}_1) + G_1 \tag{4.39}$$

$$\vdots$$

$$\bar{x}(\bar{t}_N) = F_{N-1}\bar{x}(\bar{t}_{N-1}) + G_{N-1} \tag{4.40}$$

and F_i and G_i are calculated as

$$[F_i, G_i] = c2dm(Ai, Bi, [], [], dti, `zoh`) \tag{4.41}$$

4.2.1 Constraints

As restricoes do controlador devem descrever os limites de operacao da dinamica do modelo e ainda encontrar uma solução que leve o sistema para mais proximo da referencia.

ESCREVER EQUACOES DE RESTRICAO

descrever que a trajetoria deve ser admissivel e forma com que o sistema calcula o controle tambem deve ser admissivel

fazer um rapido comentario sobre a forma que essa restricao é trabalhada no capitulo 2

4.2.2 Feasibility region

nao aplicavel ????

como calcular ??? restricoes do problem de prog quad

sobre o controle

variacoes do instante de chaveadmento

mas envolvem o estado inicial do ciclo

sea dependencia das restricoes com respeito ao estao inicial for linear

entao podemos encarar o x0 como uma variavel junto com as variaveis de controle

com isso as restricoes descrevem um polietro aumenta do de controle e estado inicial

geometria computacional mpttoolbox

projeta a geometria no subespaco

pode haver melhoria com o ponto final livre

interessante para avaliar o desempenho do controlador na regiao de factibilidade

APPLICATION EXAMPLES

- Integrador duplo(matheus)
- Patino 1
- Patino 2

4.3 Application Examples

4.3.1 Integrador duplo

4.3.2 Patino 1

4.3.3 Patino 2

Matrizes do artigo patino exemplo 2:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & -\frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}x_3(t) \end{bmatrix} \tag{4.42}$$

$$A = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & \frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \tag{4.43}$$

$$\begin{aligned}
\mathbf{u_0} &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T \\
\mathbf{u_1} &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T \\
\mathbf{u_2} &= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T \\
\mathbf{u_3} &= \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^T \\
\mathbf{u_4} &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T \\
\mathbf{u_5} &= \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T \\
\mathbf{u_6} &= \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^T \\
\mathbf{u_7} &= \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T
\end{aligned} \tag{4.44}$$

$$A_{u_0} = A\mathbf{u_0} = \begin{bmatrix} 0 \\ 0 \\ \frac{-Rx_3(t)}{L} \end{bmatrix} \tag{4.45}$$

$$A_{u_1} = A\mathbf{u_1} = \begin{bmatrix} 0 \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \tag{4.46}$$

$$A_{u_2} = A\mathbf{u_2} = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ \frac{-x_3(t)}{C_2} \\ \frac{x_2(t)-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{4.47}$$

$$A_{u_3} = A\mathbf{u_3} = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ 0 \\ \frac{E-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{4.48}$$

$$A_{u_4} = A\mathbf{u_4} = \begin{bmatrix} \frac{-x_3(t)}{C_1} \\ 0 \\ \frac{x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{4.49}$$

$$A_{u_5} = A\mathbf{u_5} = \begin{bmatrix} \frac{-x_3(t)}{C_1} \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_1(t)-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \tag{4.50}$$

$$A_{u_6} = A\mathbf{u_6} = \begin{bmatrix} 0 \\ \frac{-x_3(t)}{C_2} \\ \frac{x_2(t)-Rx_3(t)}{L} \end{bmatrix} \tag{4.51}$$

$$A_{u_7} = A\mathbf{u_7} = \begin{bmatrix} 0 \\ 0 \\ \frac{E-Rx_3(t)}{L} \end{bmatrix} \tag{4.52}$$

$$\dot{X} = A_{u0}X + B_0u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \tag{4.53}$$

$$\dot{X} = A_{u1}X + B_1u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \tag{4.54}$$

$$\dot{X} = A_{u2}X + B_2u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & -\frac{1}{C_2} \\ -\frac{1}{L} & \frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \tag{4.55}$$

$$\dot{X} = A_{u3}X + B_3u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \tag{4.56}$$

$$\dot{X} = A_{u4}X + B_4u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & 0 \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \tag{4.57}$$

$$\dot{X} = A_{u5}X + B_5u = \begin{bmatrix} 0 & 0 & -\frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ \frac{1}{L} & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \tag{4.58}$$

$$\dot{X} = A_{u6}X + B_6u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \tag{4.59}$$

$$\dot{X} = A_{u7}X + B_7u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \tag{4.60}$$

4.4 Conclusion

mostrar o modelo em malha aberta indo para a sua trajetoria ciclica estavel junto com o modelo com controle ativo mostrando a convergencia mais rapida

5 Conclusão

5.1 RASCUNHO

INICIO escrevendo aula7

=====================================

dinamica a tempo discreto

$$x(k+1) = Ax(k) + Bu(k) \tag{5.1}$$

com entrada u(k) $\in \mathbb{R}^p$ e estado $x(k) \in \mathbb{R}$ sujeitos aas seguintes restricoes:

$$u(k) \in \mathcal{U}, x(k) \in \mathcal{X}, \forall k \geq 0 \tag{5.2}$$

$$\mathcal{U} = u \in \diagdown^p : S_u u \leq b_u, \mathcal{X} = x \in \mathbb{R}^n : S_x x \leq b_x \tag{5.3}$$

deseja-se conduzir o estado para um ponto de equilibrio $\bar{x} \in int(\mathcal{X})$, associado a um valor de equilibrio $\bar{u} \in int(\mathcal{U})$.

Suponha que, sob a acao de uma dada lei de controle estabilizante da forma

$$u(k) = -K[x(k) - \bar{x}] + \bar{u} \tag{5.4}$$

o maximo conjunto admissivel quanto aas restricoes de estado e controle seja descrito por

$$\mathcal{X}_f = \{x \in \diagdown^n : S_f x \leq b_f\} \tag{5.5}$$

Se o estado inicial $x(0)$ estiver em $\mathcal{X}_f$, a lei de controle (*) podera ser aplicada a partir de $k = 0$ e o estado sera conduzido ao ponto de equilibrio desejado, sem que as restricoes sejam violadas

$\boxed{\text{O que fazer se } x(0) \notin \mathcal{X}_f \text{ ?}}$

ROTEIRO

- Conducao do estado a um conjunto alvo $\mathcal{X}_f$
- Custo terminal e custo de trajetoria
- Expressao do custoe das restricoes emprgando notacao mais concisa
- Formulacao do problema de otimizacao na forma de programacao quadratica
- Programacao quadratica: Existencia de solucao otima. Convexidade
- Solucao de problemas de programacao quadratica no Matlab

CONDUCAO DO ESTADO AO CONJUNTO $\mathcal{X}_f$

Dado um estado inicial $x(0) \notin \mathcal{X}_f$ gostariamos de obter uma sequencia de N acoes de controle de modo que

$$\begin{aligned} u(0) &\in \mathcal{U} \\ u(1) &\in \mathcal{U} \\ &\vdots \\ u(N-1) &\in \mathcal{U} \end{aligned} \tag{5.6}$$

de modo que

$$\begin{aligned} u(1) &\in \mathcal{X} \\ u(2) &\in \mathcal{X} \\ &\vdots \\ u(N-1) &\in \mathcal{X} \\ u(N) &\in \mathcal{X} \end{aligned} \tag{5.7}$$

de maneira geral, a solucao (se existir) nao sera unica. portanto, eh necessario adotar um criterio que permita escolher a melhor sequencia de acoes de controle para realizar a tarefa

para isso, empregaremos um custo da forma

$$J = \|x(N) - \bar{x}\|_{P_f}^2 + \sum_{k=0}^{N-1} \left[\|x(k) - \bar{x}\|_Q^2 + \|u(k) - \bar{u}\|_R^2 \right] \tag{5.8}$$

sendo P_f, Q, R matrizes de peso simetricas e positivo-definidas

A expressao de J envolve uma parcela de custo terminal e outra de custo de trejetoria

Consideremos inicialmente $\bar{x} = 0$ e $\bar{u} = 0$.

$$J = \|x(N)\|_{P_f}^2 + \sum_{k=0}^{N-1} \left[\|x(k)\|_Q^2 + \|u(k)\|_R^2 \right] \tag{5.9}$$

a expressao para o custo pose ser reescrita empregando a seguinte notacao:

$$\mathbf{u} = \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \in \mathbb{R}^{pN}, \mathbf{x} = \begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix} \in \mathbb{R}^{nN} \tag{5.10}$$

lembrando que o estado inicial $x(0)$ foi fixado na formulacao do problema.

com isso, a somatoria

$$\sum_{k=0}^{N-1} \|u(k)\|_R^2 = u^T(0)Ru(0) + u^t(1)Ru(1) + \cdots + u^T(N-1)Ru(N-1) \tag{5.11}$$

fica na forma

$$\underbrace{\begin{bmatrix} u^T(0) & u^T(1) & \cdots & u^T(N-1) \end{bmatrix}}_{\mathbf{u}^T} \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} \underbrace{\begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix}}_{\mathbf{u}} \tag{5.12}$$

tendo-se omitido a dimensao $(p \times p)$ das submatrizes de zeros, por brevidade.

Conventionando ainda a notacao

$$\mathbf{R} = \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} \tag{5.13}$$

ou, de forma abreviada:

$$\mathbf{R} = diag_N(R) \tag{5.14}$$

pode-se escrever

$$\sum_{k=0}^{N-1} \|u(k)\|_R^2 = \|\mathbf{u}\|_{\mathbf{R}}^2 \tag{5.15}$$

De forma similar, pode-se escrever

$$\begin{gathered} \|x(N)\|_{P_f}^2 + \sum_{k=0}^{N-1} \|x(k)\|_Q^2 = x^T(0)Qx(0) \\ +x^T(1)Qx(1) + \cdots + x^T(N-1)Qx(N-1) + x^T(N)P_f x(N) \\ = x^T(0)Qx(0) + \\ \underbrace{\begin{bmatrix} x^T(1) & x^T(2) & \cdots & x^T(N) \end{bmatrix}}_{\mathbf{x}^T} \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} \underbrace{\begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix}}_{\mathbf{x}} \end{gathered} \tag{5.16}$$

Adotando a notação

$$\mathcal{Q} = \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} \in \mathbb{R}^{nN \times nN} \tag{5.17}$$

ou, de forma abreviada:

$$\mathcal{Q} = \begin{bmatrix} diag_{N-1}(Q) & 0 \\ 0 & P_f \end{bmatrix} \tag{5.18}$$

pode-se escrever

$$\|x(N)\|_{P_f}^2 + \sum_{k=0}^{N-1} \|x(k)\|_Q^2 + \|\mathbf{x}\|_{\mathbf{Q}}^2 \tag{5.19}$$

Em resumo, o custo pode ser reescrito como

$$J(\mathbf{x}, \mathbf{u}) = \|x(0)\|_Q^2 + \|\mathbf{x}\|_{\mathbf{Q}}^2 + \|\mathbf{u}\|_{\mathbf{R}}^2 \tag{5.20}$$

Vamos agora expressar as restrições em termos de $\mathbf{x}$, $\mathbf{u}$.

As restrições podem ser escritas como

$$\mathbf{u} = \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \in \mathcal{U} \times \mathcal{U} \times \cdots \times \mathcal{U} = \mathcal{U}^N \tag{5.21}$$

$$\mathbf{x} = \begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix} \in \mathcal{X} \times \mathcal{X} \times \cdots \times \mathcal{X} = \mathcal{X}^{N-1} \times \mathcal{X}_f \tag{5.22}$$

Lembrando que

$$\mathcal{U} = \{\mathbf{u} \in \mathbb{R}^p : S_u u \leq b_u\} \tag{5.23}$$

a restrição sobre $\mathbf{u}$ pode ser expressa como

$$\begin{bmatrix} S_u & 0 & \cdots & 0 \\ 0 & S_u & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_u \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \leq \begin{bmatrix} b_u \\ b_u \\ \vdots \\ b_u \end{bmatrix} \tag{5.24}$$

ou, de forma mais concisa:

$$\mathbf{S_u u} \leq \mathbf{b_u} \tag{5.25}$$

sendo

$$\mathbf{S_u} = diag_N(S_u), \mathbf{b_u} = col_N(b_u) \tag{5.26}$$

De forma similar, lembrando que

$$\mathcal{X} = \{x \in \mathbb{R}^n : S_x x \leq b_x\}, \mathcal{X}_f = \{x \in \mathbb{R}^n : S_f x \leq b_f\} \tag{5.27}$$

ou, de forma mais concisa

$$\mathbf{S_x x} \leq \mathbf{b_x} \tag{5.28}$$

sendo

$$\mathbf{S_x} = \begin{bmatrix} diag_{N-1}(S_x) & 0 \\ 0 & S_f \end{bmatrix}, \mathbf{b_x} = \begin{bmatrix} col_{N-1}(b_x) \\ b_f \end{bmatrix} \tag{5.29}$$

No caso geral em que $\bar{x} \neq 0, \bar{u} \neq 0$

$$J = \|x(N) - \bar{x}\|^2_{P_f} + \sum_{k=0}^{N-1} \left[\|x(k) - \bar{x}\|^2_Q + \|u(k) - \bar{u}\|^2_R\right] \tag{5.30}$$

torna-se

$$J(\mathbf{x}, \mathbf{u}) = \|x(0) - \bar{x}\|^2_{P_f} + \|\mathbf{x} - col_N(\bar{x})\|^2_{\mathbf{Q}} + \|\mathbf{u} - col_N(\bar{u})\|^2_{\mathbf{R}} \tag{5.31}$$

O problema de otimização a ser resolvido pode ser expresso como

$$\min_{\mathbf{x} \in \mathbb{R}^{nN}, \mathbf{u} \in \mathbb{R}^{pN}} J(x, u) \tag{5.32}$$

sendo

$$J(\mathbf{x}, \mathbf{u}) = \|x(0) - \bar{x}\|^2_Q + \|\mathbf{x} - col_N(\bar{x})\|^2_{\mathbf{Q}} + \|\mathbf{u} - col_N(\bar{u})\|^2_{\mathbf{R}} \tag{5.33}$$

com $\mathbf{u}$, $\mathbf{u}$ sujeitas a

$$\mathbf{S_x x} \leq \mathbf{b_x} \tag{5.34}$$

$$\mathbf{S_u u} \leq \mathbf{b_u} \tag{5.35}$$

Vamos expressar o vinculo entre os vetores

$$\mathbf{u} = \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix} \tag{5.36}$$

com base na equação de estado

$$x(k+1) = Ax(k) + Bu(k) \tag{5.37}$$

e na condição inicial $x(0)$.

Neste ponto, vale lembrar a solução da equação de estado:

$$x(k) = A^k x(0) + \sum_{i=0}^{k-1} A^{k-1-i} Bu(i) \tag{5.38}$$

que pose ser escrita para $k = 1, 2, ..., N$ como

$$x(1) = Ax(0) + Bu(0) \tag{5.39}$$

$$x(2) = A^2 x(0) + ABu(0) + Bu(1) \tag{5.40}$$

$$\vdots \tag{5.41}$$

$$x(N) = A^N x(0) A^{N-1} Bu(0) + \cdots + ABu(N-2) + Bu(N-1) \tag{5.42}$$

ou ainda

$$\begin{bmatrix} x(1) \\ x(2) \\ \vdots \\ x(N) \end{bmatrix} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} x(0) + \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \tag{5.43}$$

De forma mais concisa, vamos escrever

$$\mathbf{x} = \mathbf{A}x(0) + \mathbf{B}\mathbf{u} \tag{5.44}$$

com

$$\mathbf{A} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}, \mathbf{B} = \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \tag{5.45}$$

Tendo em vista o vínculo

$$\mathbf{x} = \mathbf{A}x(0) + \mathbf{B}\mathbf{u} \tag{5.46}$$

pode-se reescrever a função de custo como

$$\begin{aligned} J(\mathbf{x},\mathbf{u}) &= \|x(0) - \bar{x}\|_Q^2 + \|\mathbf{x} - col_N(\bar{x})\|_{\mathbf{Q}}^2 + \|\mathbf{u} - col_N(\bar{u})\|_{\mathbf{R}}^2 \\ &= \|x(0) - \bar{x}\|_Q^2 \\ &+ \left[\mathbf{A}x(0) + \mathbf{B}\mathbf{u} - col_N(\bar{x})\right]^T \mathbf{Q}\left[\mathbf{A}x(0) + \mathbf{B}\mathbf{u} - col_N(\bar{x})\right] \\ &+ \left[\mathbf{u} - col_N(\bar{u})\right]^T \mathbf{R}\left[\mathbf{u} - col_N(\bar{u})\right] \\ J(\mathbf{x},\mathbf{u}) &= \|x(0) - \bar{x}\|_Q^2 \\ &+ \left[\mathbf{A}x(0) - col_N(\bar{x})\mathbf{B}\mathbf{u}\right]^T \mathbf{Q}\left[\mathbf{A}x(0) - col_N(\bar{x})\mathbf{B}\mathbf{u}\right] \\ &+ \mathbf{u}^T\mathbf{R}\mathbf{u} - \mathbf{u}^T\mathbf{R}\left[col_N(\bar{u})\right] - \left[col_N(\bar{u})\right]^T \mathbf{R}\mathbf{u} + \|col_N(\bar{u})\|_{\mathbf{R}}^2 \end{aligned} \tag{5.47}$$

$$\begin{aligned} J(\mathbf{x},\mathbf{u}) &= \|x(0) - \bar{x}\|_Q^2 + + \|col_N(\bar{u})\|_{\mathbf{R}}^2 \\ &\left[\mathbf{A}x(0) - col_N(\bar{x})\right]^T \mathbf{Q}\mathbf{B}\mathbf{u} + y \\ &- \mathbf{u}^T\mathbf{R}\left[col_N(\bar{u})\right] - \left[col_N(\bar{u})\right]^T \mathbf{R}\mathbf{u} + \mathbf{u}^T(\mathbf{R} + \mathbf{B}^T\mathbf{Q}\mathbf{B})\mathbf{u} \end{aligned} \tag{5.48}$$

Observacoes:

O termo constante será denotado por c_0

Lembrando que a matriz $\mathbf{Q}$ é simétrica, tem-se

$$\left[\mathbf{A}x(0) - col_N(\bar{x})\right]^T \mathbf{Q}\mathbf{B}\mathbf{u} = \left\{\mathbf{u}^T\mathbf{B}^T\mathbf{Q}\left[\mathbf{A}x(0) - col_N(\bar{x})\right]\right\}^T \tag{5.49}$$

Como esses termos são escalares, conclui-se que são iguais entre si. Logo:

$$\begin{aligned} &\left[\mathbf{A}x(0) - col_N(\bar{x})\right]^T \mathbf{Q}\mathbf{B}\mathbf{u} + \mathbf{u}^T\mathbf{B}^T\mathbf{Q}\left[\mathbf{A}x(0) - col_N(\bar{x})\right] \\ &= 2\left[\mathbf{A}x(0) - col_N(\bar{x})\right]^T \mathbf{Q}\mathbf{B}\mathbf{u} \end{aligned} \tag{5.50}$$

De forma similar:

$$-\mathbf{u}^T\mathbf{R}\left[col_N(\bar{u})\right] - \left[col_N(\bar{u})\right]^T\mathbf{R}\mathbf{u} = -2\left[col_N(\bar{u})\right]^T\mathbf{R}\mathbf{u} \tag{5.51}$$

Finalmente, pode-se escrever

$$\begin{gathered} J(\mathbf{u}) = \mathbf{u}^T(\mathbf{B}^T\mathbf{Q}\mathbf{B} + \mathbf{R})\mathbf{u} \\ +2\left\{\left[\mathbf{A}x(0) - col_N(\bar{x})\right]^T\mathbf{Q}\mathbf{B} - \left[col_N(\bar{u})\right]^T\mathbf{R}\right\}\mathbf{u} = c_0 \end{gathered} \tag{5.52}$$

com

$$c_0 = \|x(0) - \bar{x}\|_Q^2 + \|\mathbf{A}x(0) - col_N(\bar{x})\|_{\mathbf{R}}^2 + \|col_N(\bar{u})\|_{\mathbf{R}}^2 \tag{5.53}$$

Note-se que passaremos a usar a notação $J(\mathbf{u})$ em lugar de $J(\mathbf{x}, \mathbf{u})$, uma vez que $\mathbf{x}$ foi eliminada da expressão do custo.

Para terminar a reformulação do problema de otimização em termos de $\mathbf{u}$, deve-se considerar o vínculo entre $\mathbf{x}$ e $\mathbf{u}$ nas **restrições**

Considerando novamente o vínculo

$$\mathbf{x} = \mathbf{A}x(0) + \mathbf{B}\mathbf{u} \tag{5.54}$$

a restrição

$$\mathbf{S_x}\mathbf{x} \le \mathbf{b_x} \tag{5.55}$$

pode ser reformulada como

$$\mathbf{S_x}\mathbf{B}\mathbf{u} \le \mathbf{b_x} - \mathbf{S_x}\mathbf{A}x(0) \tag{5.56}$$

Em conjunto com $\mathbf{S_u}\mathbf{u} \le \mathbf{b_u}$, pode-se então expressar as restrições na forma

$$\begin{bmatrix} \mathbf{S_x}\mathbf{B} \\ \mathbf{S_u} \end{bmatrix}\mathbf{u} \le \begin{bmatrix} \mathbf{b_x} - \mathbf{S_x}\mathbf{A}x(0) \\ \mathbf{b_u} \end{bmatrix} \tag{5.57}$$

ou, de maneira mais sucinta:

$$\mathbf{S}\mathbf{u} \le \mathbf{b} \tag{5.58}$$

com

$$\mathbf{S} = \begin{bmatrix} \mathbf{S_x B} \\ \mathbf{S_u} \end{bmatrix}, \mathbf{b} = \begin{bmatrix} \mathbf{b_x} - \mathbf{S_x A} x(0) \\ \mathbf{b_u} \end{bmatrix} \tag{5.59}$$

Em resumo, o problema de otimização passa a ser expresso como

$$\min_{\mathbf{u} \in ^{pN}} J(\mathbf{u}) s.a. \mathbf{S u} \leq \mathbf{b} \tag{5.60}$$

sendo

$$\begin{gathered} J(\mathbf{u}) \mathbf{u^T} (\mathbf{B}^T \mathbf{Q B} + \mathbf{R}) \mathbf{u} \\ +2 \Big\{ [\mathbf{A} x(0) - col_N(\bar{x})]^T \mathbf{Q B} - [col_N(\bar{u})]^T \mathbf{R} \Big\} \mathbf{u} + c_0 \end{gathered} \tag{5.61}$$

Trata-se da minimização de um **custo quadrático** sujeito a **restrições lineares**:

PROBLEMA DE PROGRAMACAO QUADATICA

Para facilitar a análise do problema de programação quadrática, passaremos a escrever o custo

$$\begin{gathered} J(\mathbf{u}) = \mathbf{u}^T (\mathbf{B}^T \mathbf{Q B} + \mathbf{R}) \mathbf{u} + \\ 2 \Big\{ [\mathbf{A} x(0) - col_N(\bar{x})]^T \mathbf{Q B} - [col_N(\bar{u})]^T \mathbf{R} \Big\} \mathbf{u} + c_0 \end{gathered} \tag{5.62}$$

nesta forma padronizada:

$$J(\mathbf{u}) = \frac{1}{2} \mathbf{u}^T J_{uu} \mathbf{u} f^T \mathbf{u} + c_0 \tag{5.63}$$

com

$$J(\mathbf{u}) = 2(\mathbf{B}^T \mathbf{Q B} + \mathbf{R}), f = 2 \Big\{ \mathbf{B}^T \mathbf{Q} [\mathbf{A} x(0) - col_N(\bar{x})] - \mathbf{R} col_N(\bar{u}) \Big\} \tag{5.64}$$

lembrando que $\mathbf{Q}$ e $\mathbf{R}$ são simétricas.

TERMINOLOGIA E NOTACAO

$$J(\mathbf{u}) \frac{1}{2} \mathbf{u}^T J_{\mathbf{uu}u} + f^T \mathbf{u} = c_0 \tag{5.65}$$

1) A matriz $J_{\mathbf{uu}}$ corresponde à chamada **Hessiana** do custo, que é definica como

$$J_{\mathbf{uu}} = \begin{bmatrix} \frac{\partial^2 J}{\partial u_1^2} & \frac{\partial^2 J}{\partial u_1 \partial u_2} & \cdots & \frac{\partial^2 J}{\partial u_1 \partial u_N} \\ \frac{\partial^2 J}{\partial u_2 \partial u_1} & \frac{\partial^2 J}{\partial u_2^2} & \cdots & \frac{\partial^2 J}{\partial u_2 \partial u_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 J}{\partial u_N \partial u_1} & \frac{\partial^2 J}{\partial u_N \partial u_2} & \cdots & \frac{\partial^2 J}{\partial u_N^2} \end{bmatrix} \tag{5.66}$$

sendo $u_1, u_2, \ldots, u_N$ como componentes de $\mathbf{u}$. Vide **material suplementar 1** ao final dos slides para mais detalhes.

Vale notar que $J_{\mathbf{uu}} = 2(\mathbf{B}^T\mathbf{Q}\mathbf{B} + \mathbf{R})$ é simétrica e positivo definida, pois $\mathbf{Q}$ e $\mathbf{R}$ são simétricas e positivo definidas.

5.1.1 FIM

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Appendix A - Tópicos de Dilema Linear

A.1 Modelo de simulação

```
+engine
 ├── CORES.m
 ├── fun_custo_patino.m
 ├── get_config_sim_matheus.m
 ├── get_config_sim_patino_1.m
 ├── get_config_sim_patino_2.m
 ├── get_config.m
 ├── get_otmin_opt.m
 ├── get_ts.m
 ├── get_x0.m
 ├── get_xr.m
 ├── nonlincon_patino.m
 ├── otmin.m
 ├── sim_1.m
 └── sim_n.m
+mpc
 ├── construcao_modelo_instantes.m
 ├── matrizes_ss_mpc_dualmode_switching.m
 └── mpc_dualmode_switching.m
```

A.1.1 get_x0

funcao de calculo da condicao inicial

A.1.2 get_xf

Texto do modelo de simulação sendo colocado aqui.

Annex A - Exemplo de um Primeiro Anexo

A.1 Uma Seção do Primeiro Anexo

Algum texto na primeira seção do primeiro anexo.

FOLHA DE REGISTRO DO DOCUMENTO

1. CLASSIFICAÇÃO/TIPO	2. DATA	3. DOCUMENTO Nº	4. Nº DE PÁGINAS
TD	25 de março de 2015	DCTA/ITA/DM-018/2015	56

5. TÍTULO E SUBTÍTULO:

Robust control of linear systems with Switched Actuators Subjected to Dwell-Time Constraints

6. AUTOR(ES):

Daniel Vieira

7. INSTITUIÇÃO(ÕES)/ÓRGÃO(S) INTERNO(S)/DIVISÃO(ÕES):

Instituto Tecnológico de Aeronáutica – ITA

8. PALAVRAS-CHAVE SUGERIDAS PELO AUTOR:

Cupim; Cimento; Estruturas

9. PALAVRAS-CHAVE RESULTANTES DE INDEXAÇÃO:

Cupim; Dilema; Construção

10. APRESENTAÇÃO: (**X**) **Nacional** ( ) **Internacional**

ITA, São José dos Campos. Curso de Mestrado. Programa de Pós-Graduação em Engenharia Aeronáutica e Mecânica. Área de Sistemas Aeroespaciais e Mecatrônica. Orientador: Prof. Dr. Adalberto Santos Dupont. Coorientadora: Profª. Drª. Doralice Serra. Defesa em 05/03/2015. Publicada em 25/03/2015.

11. RESUMO:

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o systema à trajetória cíclica alvo.

12. GRAU DE SIGILO:

(**X**) **OSTENSIVO** ( ) **RESERVADO** ( ) **SECRETO**